Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	12-10-2026
Team ID	SWUID2026403484
Project Name	HotelPro - Hotel Room Booking and Management System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Product Backlog & Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint 1	Project Setup & Auth	USN 1	As a developer, I can set up the React + Node.js/Express project, MongoDB connection, and Git repository (NFR-MNT-001/003).	8	High	
Sprint 1	Project Setup & Auth	USN 2	As a visitor, I can register with name, email, phone number and password, validated for uniqueness and strength (FR-REG-01, 02).	5	High	
Sprint 1	Project Setup & Auth	USN 3	As a system, I shall hash passwords securely using bcrypt before storing them in MongoDB (FR-REG-03).	3	High	
Sprint 1	Project Setup & Auth	USN 4	As a registered user, I can log in and out securely with JWT session handling and role-based access control (FR-REG-04 to 06, FR-JWT-01 to 05).	8	High	
Sprint 1	User Profile	USN 5	As a user, I can view and update my profile details and manage my account settings (FR-REG-07, 08).	3	Medium	
Sprint 2	Hotel & Room Management	USN 6	As an admin, I can add a hotel with name, address, city, description, amenities and images	5	High	

			(FR-HRM-01).			
Sprint 2	Hotel & Room Management	USN 7	As an admin, I can edit, update or delete hotel details (FR-HRM-02).	3	High	
Sprint 2	Hotel & Room Management	USN 8	As an admin, I can add rooms to a hotel with room number, type (Single, Double, Deluxe, Suite), price, capacity, amenities and status (FR-HRM-03).	8	High	
Sprint 2	Hotel & Room Management	USN 9	As an admin, I can update room status (Available, Occupied, Maintenance, Inactive) and prevent deletion of rooms with active bookings (FR-HRM-04, 05).	5	High	

Sprint	**Functional Requirement (Epic)**	**User Story Number**	**User Story / Task**	**Story Points**	**Priority**	**Team Members**
Sprint 2	User & Role Management	USN 10	As an admin, I can manage user accounts, change roles (customer, staff, admin) and activate or deactivate accounts (FR-ADM-03).	5	High	
Sprint 3	Search & Availability	USN 11	As a customer, I can search rooms by city/hotel, check-in date, check-out date and number of guests with date validation (FR-SRC-01, 05).	8	High	
Sprint 3	Search & Availability	USN 12	As a customer, I can filter and sort search results by price, room type, amenities and rating, returning in under 2 seconds (FR-SRC-02, 06).	5	High	
Sprint 3	Search & Availability	USN 13	As a customer, I can view detailed room information including images, price per night, capacity and amenities (FR-SRC-04).	3	High	

Sprint 3	Room Booking	USN 14	As an authenticated customer, I can book an available room for selected dates with automatic total amount calculation including taxes (FR-BKG-01, 02).	8	High	
Sprint 3	Room Booking	USN 15	As a system, I shall re-verify room availability at confirmation and generate a unique booking ID with Pending status (FR-BKG-03, 05).	5	High	
Sprint 4	Booking Management	USN 16	As a customer, I can view my current, upcoming and past bookings in My Bookings (FR-BMG-01).	3	High	
Sprint 4	Booking Management	USN 17	As a customer, I can cancel or modify my booking dates subject to policy, making cancelled rooms available again (FR-BMG-02, 03, 06).	8	High	
Sprint 4	Booking Management	USN 18	As staff and admins, I can view, search and filter all bookings by status, date or guest across the system (FR-BMG-04).	5	High	
Sprint 4	Payment Processing	USN 19	As a customer, I can pay for bookings securely via payment gateway (cards, UPI, net banking) without sensitive card data storage (FR-PAY-01, 02).	8	High	
Sprint 4	Payment Processing	USN 20	As a system, I record transaction details, update booking status to Confirmed upon success, generate receipts and handle offline counter payments (FR-PAY-03, 04, 06, 07).	5	High	
Sprint 5	Guest Management	USN 21	As hotel staff, I can add, view, edit and search guest records with contact details and ID proof types/numbers (FR-GST-01).	5	High	

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Sprint 5	Guest Management	USN 22	As hotel staff, I can link guest records to bookings, record accompanying guests and restrict data access to authorized staff (FR-GST-02, 03, 04).	5	High	
Sprint 5	Check-in & Check-out	USN 23	As hotel staff, I can check in guests against confirmed bookings, update booking status to Checked-in and set room to Occupied (FR-CIO-01, 02).	8	High	
Sprint 5	Check-in & Check-out	USN 24	As hotel staff, I can check out guests, verify payment completion, record actual timestamps and set room to Available or Maintenance (FR-CIO-03, 04, 05).	8	High	
Sprint 6	Admin Dashboard	USN 25	As an admin, I can view key metrics (total bookings, active bookings, cancellations, occupancy rate, available rooms, total revenue) (FR-ADM-01).	5	Medium	
Sprint 6	Admin Dashboard	USN 26	As an admin, I can view summaries by day, week, month and view all bookings and payment transaction logs (FR-ADM-02, 04).	5	Medium	
Sprint 6	Admin Dashboard	USN 27	As an admin, I can export reports in CSV or PDF format for business analysis (FR-ADM-05).	3	Low	
Sprint 6	Testing & Deployment	USN 28	As a team, we complete API testing with Postman, security audits, responsive UI verification across browsers and deployment (Section 4-6).	8	High	

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	27	14 Days	19 Oct 2026	1 Nov 2026		

Sprint-2	29	14 Days	2 Nov 2026	15 Nov 2026		
Sprint-3	30	14 Days	16 Nov 2026	29 Nov 2026		
Sprint-4	31	14 Days	30 Nov 2026	13 Dec 2026		
Sprint-5	31	14 Days	14 Dec 2026	27 Dec 2026		
Sprint-6	29	14 Days	28 Dec 2026	10 Jan 2027		
Total	**177**	**84 Days**	19 Oct 2026	10 Jan 2027		

Velocity

Total story points: 177 across 6 sprints of 14 days (84 days). Average velocity = 177 / 6 = 29.5 story points per sprint (about 2.11 points per day).

Priority key: High = must-have for the booking flow; Medium = should-have; Low = could-have. Story points use a 1-8 scale. Actual completion and release dates are filled in as each sprint closes.

Burndown Chart

